

Multinomial classification of Alzheimer's disease based on augmented T1-weighted magnetic resonance imaging using a convolutional neural network

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Introduction/Background

Alzheimer's disease (AD) is a progressive brain disease that is the most common cause of dementia in old age. Characterized by amnesia, emotional lability, and a decline in language capabilities, AD leads to the loss of the ability to sustain basic bodily functions. Although AD has become one of the most expensive diseases to treat, it is poorly understood and only diagnosed through autopsies. Diagnostic factors include medical history and neuropsychological assessment. Noninvasive methods for diagnosis are necessary for early detection. AD is associated with a loss of neurons in the cerebral cortex, and an accumulation of amyloid beta plaques and neurofibrillary tangles. Magnetic resonance imaging (MRI) allows for the identification of AD signs such as brain atrophy, ventricular enlargement, and white matter hyperintensities. Studies have employed supervised learning methods such as logistic regression and support vector machines to diagnose AD from brain scans. Bae et al. achieved a testing accuracy of 83% and an area under the curve (AUC) of 0.88 with a convolutional neural network (CNN), a promising sign for artificial intelligence methods for AD diagnosis.

Research Objectives

Our goal was to improve the accuracy and AUC of a deep learning model for the classification of MRI scans into different degrees of AD, to use various methods to improve the generalization of the model for applications in real-world clinical settings, and to optimize the hyperparameters of the model to ensure robust performance.

Methodology

We used a CNN to classify AD severity based on MRI axial brain scans. We used the Open Access Series of Imaging Studies (OASIS) dataset of 6,408 preprocessed MRI scans sorted into four Clinical Dementia Rating-based categories: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented. Images were standardized by converting color space from BGR to RGB and resized from 176×208 to 224×224 pixels to facilitate convolution operations. The dataset was split into 80% for training and validation and 20% for testing. To improve model generalization, image augmentation techniques such as rotation, affine transformations, and contrast and brightness jittering were applied to a 10% sample of training and validation data, which was added to the dataset. 80% of this processed dataset was used for training, and the remaining 20% for validation. The CNN architecture included convolutional layers with pooling and batch normalization, ReLU activations, and dropout layers. A final softmax layer produced class probability distributions. Weighted cross-entropy loss guided model training. Two optimizers, Adam and stochastic gradient descent with momentum, were used. Hyperparameters learning rate, batch size, epoch number, and optimizer type were tuned using Bayesian optimization through the Weights & Biases platform. Performance was assessed using accuracy and multiclass AUC.

Results

Out of 25 runs, the model with the highest performance produced a 87.53% accuracy overall over all four classes: 93.51% for NonDemented images, 79.35 for VeryMildDemented, 87.29 for MildDemented, and 78.32 for ModerateDemented. The hyperparameters for this model were batch size 32, 70 training epochs, a learning rate of 3.333×10^{-5} , and the Adam optimizer.

Conclusions/Implications

Our model improved upon previous studies' performance with a higher testing accuracy, even compared to binary classifiers. Furthermore, for each class, the testing accuracy was within five to 10% to the average accuracy over all four classes, indicating that the model classified all severities of AD with a relatively high accuracy. The performance of our model reaffirms past studies which have suggested CNNs achieve high performance on MRI imaging for diagnosis, and builds upon those studies by showing potential for CNNs in AD severity classification. The testing accuracy of our model indicates potential applicability to unseen data and real-life applications. A refined, improved model may be used in clinical settings for the diagnosis and classification of AD in patients, perhaps incorporated into clinical-level workflow as a decision support tool for doctors. Future studies should expand the number of levels of severities of AD for classification. Furthermore, a larger training dataset with non-OASIS images will allow for greater generalization. We propose our model as a foundational step toward developing a multimodal machine learning framework. By integrating additional factors such as age and medical history with MRI, future models may achieve improved performance, as the combination of modalities may enhance versatility and enable more accurate classification.